

Tutorial 3 — Entity-Relationship Modeling

BCL1223 / BIT1223 DATABASE SYSTEMS

Student Name: Chan Jing Yi

Student ID: SUOL2500321

Course: BCL1223 Database
Fundamentals

Date: July 10, 2026

Question 1: What is ERD?

An Entity-Relationship Diagram (ERD) is a foundational conceptual data modeling tool that provides a graphical representation of the logical structure of a database. Originally introduced by Chen (1976), the model represents the real-world operational domain of an enterprise in terms of three primary semantic abstractions: **entities** (which represent unique physical or abstract objects of interest), **attributes** (which describe the properties or characteristics of those objects), and **relationships** (which define the logical connections or associations between entity types). By decoupling logical structure from physical implementation constraints, the ERD allows database designers to focus entirely on data semantics, business rules, and constraints.

The ERD serves as an essential communication mechanism throughout the software development lifecycle, bridging the gap between database designers, system analysts, and non-technical business stakeholders. It provides a visual language to validate requirements, ensuring that the conceptual structure of the database accurately reflects business workflows before any database implementation begins. Ultimately, the conceptual ER model is mapped to a logical model (such as the relational model) and subsequently defined within the Database Management System (DBMS) system catalog, enforcing semantic constraints and data integrity at the system level (Elmasri & Navathe, 2017).

Question 2: List FOUR types of relationship in ERD.

Relationships in Entity-Relationship modeling describe how instances of different entity types are logically linked. They are classified based on their

cardinality ratios and structural degree. Four major types of relationships include:

- **One-to-One (1:1) Relationship:** Occurs when a single instance of Entity A is associated with at most one instance of Entity B, and conversely, a single instance of Entity B is associated with at most one instance of Entity A. For example, a physical retail store and its manager represent a 1:1 relationship; a store has exactly one manager, and an employee can manage at most one store.
- **One-to-Many (1:M) Relationship:** Occurs when a single instance of Entity A can be associated with multiple instances of Entity B, but each instance of Entity B is linked to exactly one instance of Entity A. This is the most common structural relationship in relational database design. For example, a department employs many staff members, but each staff member belongs to a single department.
- **Many-to-Many (M:N) Relationship:** Occurs when an instance of Entity A can be associated with multiple instances of Entity B, and an instance of Entity B can simultaneously be associated with multiple instances of Entity A. For example, a student can enroll in multiple courses, and each course contains multiple students. In physical relational databases, \$M:N\$ relationships are resolved into two \$1:M\$ relationships using an associative (junction) entity.
- **Unary (Recursive) Relationship:** Occurs when a relationship exists between instances of the same entity type, where the entity type participates in the relationship in multiple roles. For example, in an organizational hierarchy, an employee can supervise other employees; both the supervisor and the supervisee are instances of the same EMPLOYEE entity.

Question 3: List FIVE entity for restaurant.

To represent the operational environment of a restaurant, five core entities are identified below, outlining their conceptual role and candidate identifiers:

1. **CUSTOMER:** Tracks the details of patrons who dine at the restaurant, make reservations, or place orders.
 - *Primary Identifier:* CustomerID
 - *Key Attributes:* CustomerName, PhoneNumber, EmailAddress, LoyaltyStatus
2. **STAFF:** Tracks the restaurant's employees, including chefs, waitstaff, hosts, and managers, enabling labor tracking and scheduling.

- *Primary Identifier:* StaffID
 - *Key Attributes:* StaffName, Role, HourlyRate, ContactNumber
3. **MENU_ITEM:** Represents individual food and beverage products offered by the restaurant, including pricing and category classifications.
- *Primary Identifier:* MenuItemID
 - *Key Attributes:* ItemName, UnitPrice, Category (e.g., Appetizer, Main, Dessert, Drink), IsAvailable
4. **DINING_TABLE:** Represents physical seating tables within the dining area, allowing table management and seating arrangements.
- *Primary Identifier:* TableNumber
 - *Key Attributes:* SeatingCapacity, SectionLocation (e.g., Main Room, Patio, Bar), CurrentStatus
5. **CUSTOMER_ORDER:** Represents a specific transaction where a customer purchases food or drinks. It binds customers, staff, tables, and purchased items.
- *Primary Identifier:* OrderID
 - *Key Attributes:* OrderTimestamp, TotalAmount, PaymentMethod, TableNumber (FK), CustomerID (FK)

Question 4: List FIVE attributes for BUILDING, CAR and T-SHIRT.

Attributes represent the properties or characteristics of an entity type. Five distinct attributes, along with their data types and primary key (PK) designations, are specified for each entity:

1. BUILDING Entity

Attribute Name	Data Type	Description & Constraints
BuildingID (PK)	Alphanumeric	Unique identifier code for each building.
BuildingName	String	Descriptive name of the building (NOT NULL).
PhysicalAddress	String	Full street address representing building location.
NumberOfFloors	Integer	The number of stories, indicating height (NOT NULL, > 0).
TotalUsableArea	Decimal	The total internal square footage or square meters.

2. CAR Entity

Attribute Name	Data Type	Description & Constraints
VIN (PK)	Alphanumeric	Standard 17-character unique Vehicle Identification Number.
LicensePlateNumber	String	Registration plate number issued by the state (Unique).
Make	String	The manufacturer of the vehicle (e.g., Honda, Ford).
Model	String	The specific product name (e.g., Civic, Mustang).
YearOfManufacture	Integer	The year of assembly (NOT NULL, 1886 to current).

3. T-SHIRT Entity

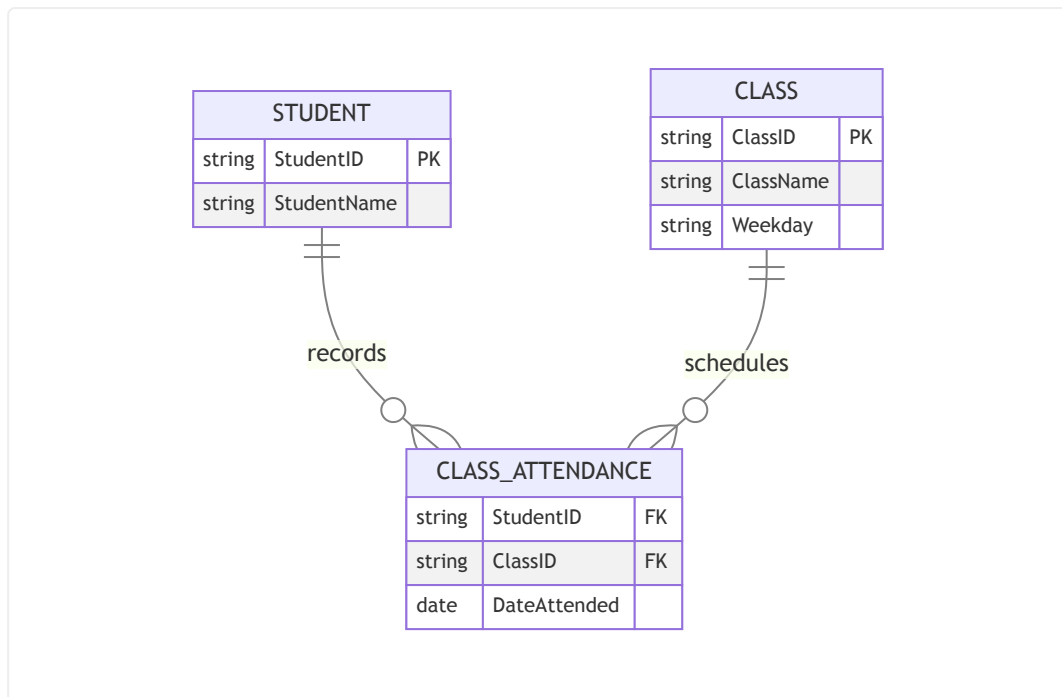
Attribute Name	Data Type	Description & Constraints
SKU (PK)	Alphanumeric	Unique Stock Keeping Unit code for inventory.
SizeCode	String	Garment size (NOT NULL, limited to XS, S, M, L, XL, XXL).
ColorName	String	Color name of the T-shirt fabric (e.g., Navy, White).
FabricComposition	String	Material breakdown (e.g., "100% Organic Cotton").
UnitPrice	Decimal	Retail price (NOT NULL, scale of 2).

Question 5: Identify the relationship and draw the relationship for the following situation:

-a A student attend many classes on weekdays.

Relationship Type: Many-to-Many (M:N)

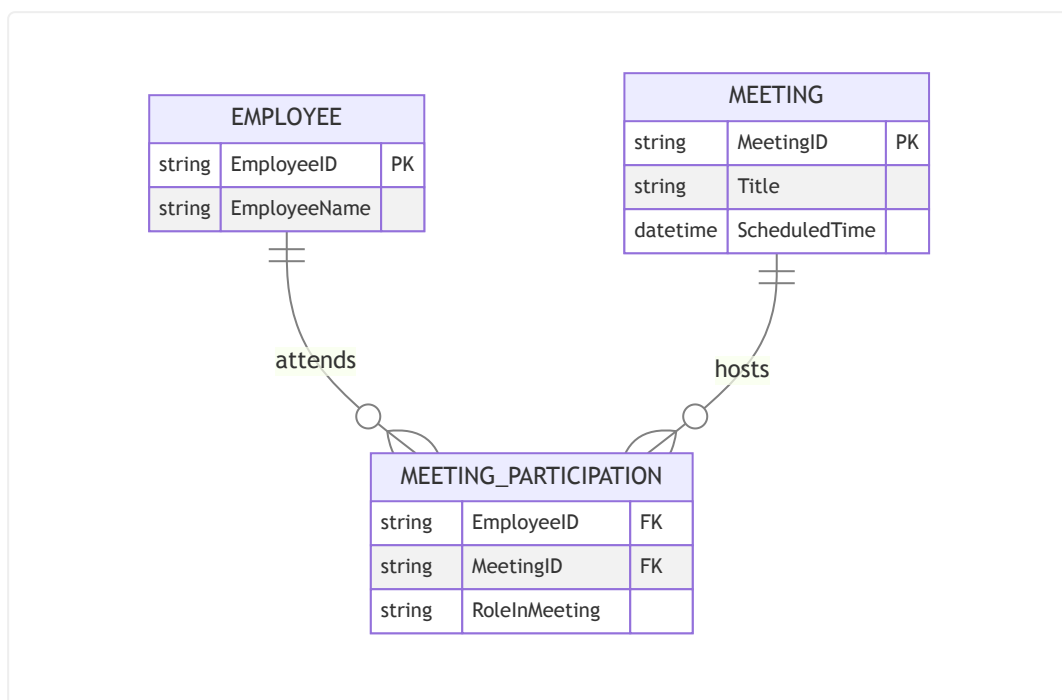
Conceptual Analysis: While the statement specifies that a single student attends multiple classes, realistic data modeling dictates that each class is also attended by multiple students. Because this represents an \$M:N\$ relationship, it is resolved in the relational model using an associative entity, CLASS_ATTENDANCE, which acts as a bridge containing composite foreign keys.



-b Many employees has join the company meeting.

Relationship Type: Many-to-Many (M:N)

Conceptual Analysis: Multiple employees join corporate meetings. Since an employee attends many meetings and a meeting hosts many employees, the relation is Many-to-Many. The relationship is decomposed by creating the associative entity **MEETING_PARTICIPATION**.

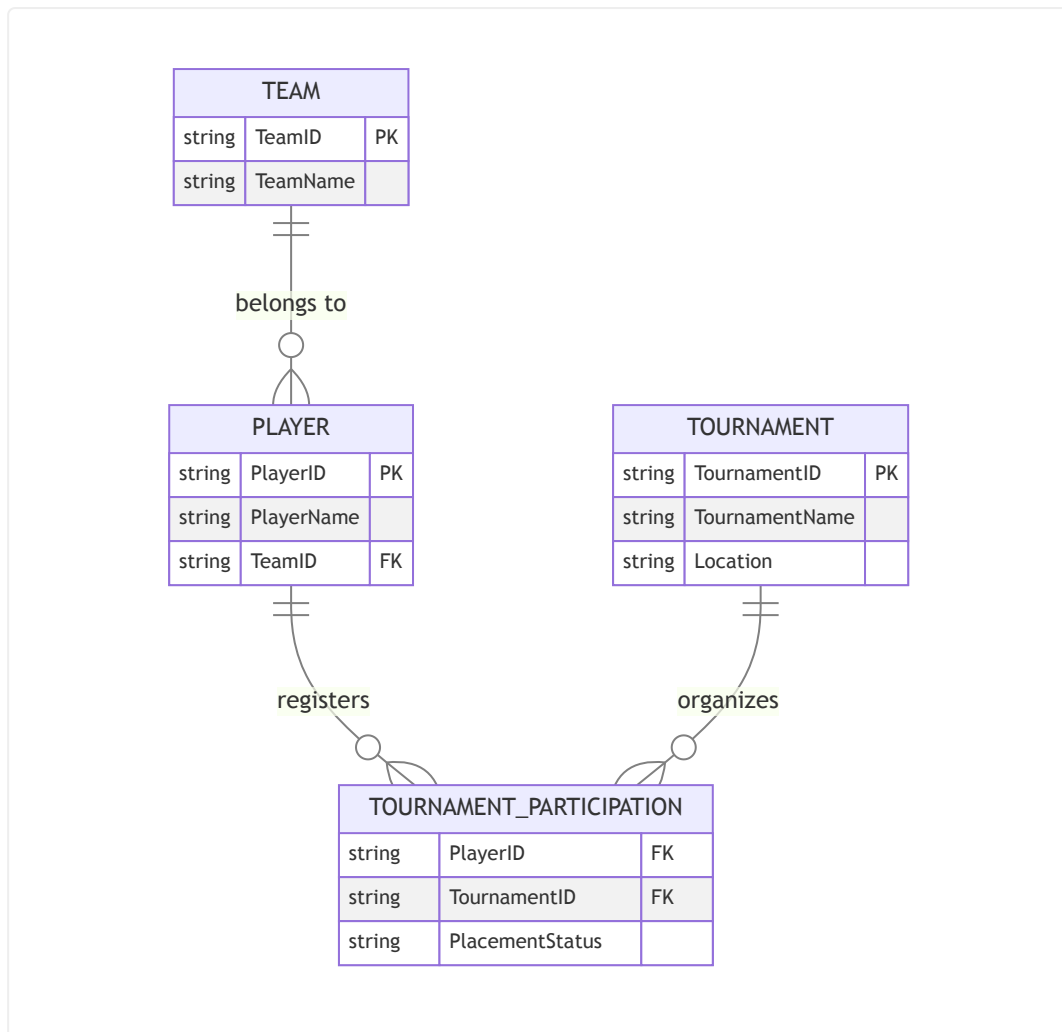


-c Many players in the Red Team have joined many tournament around the world.

Relationship Type: Many-to-Many (M:N)

Conceptual Analysis: Individual players participate in multiple

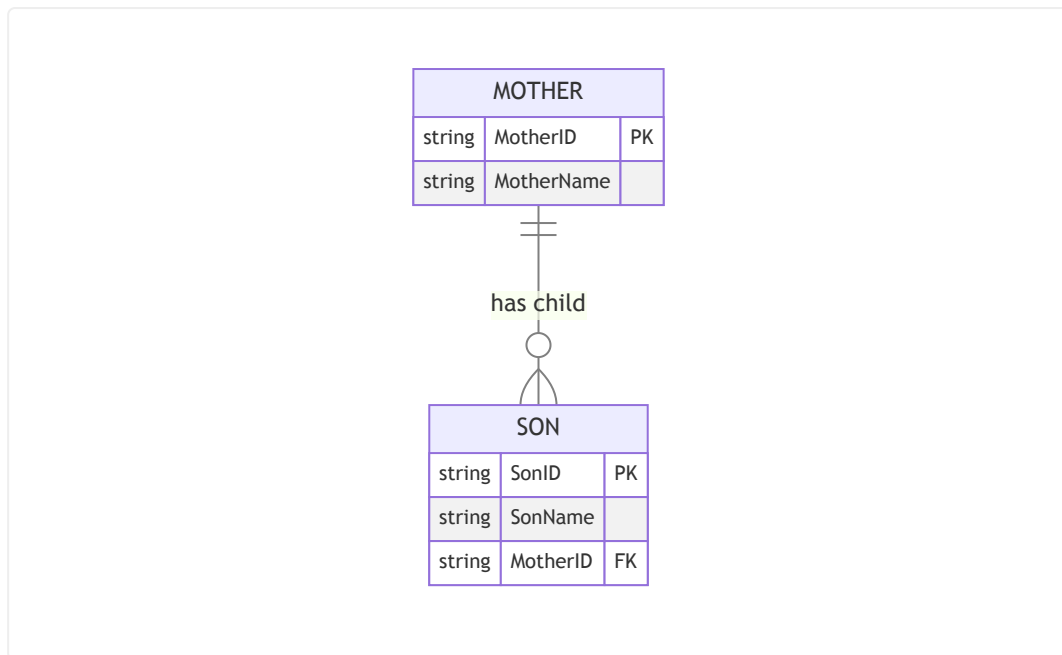
tournaments, and a tournament hosts multiple players. This creates an \$M:N\$ relationship between PLAYER and TOURNAMENT, resolved by the associative entity TOURNAMENT_PARTICIPATION. The phrase "Red Team" indicates a parent \$1:M\$ relationship where a team contains many players.



-d The mother has prepared breakfast to her son.

Relationship Type: One-to-Many (1:M) or Ternary

Conceptual Analysis: Family structures dictate that a mother can have multiple biological children (including sons), but a son has exactly one biological mother, representing a \$1:M\$ relationship between MOTHER and SON. In a transaction database tracking daily activities, this event can also be represented as a ternary association between MOTHER, SON, and MEAL (Breakfast). The binary parent-child relationship is modeled below.



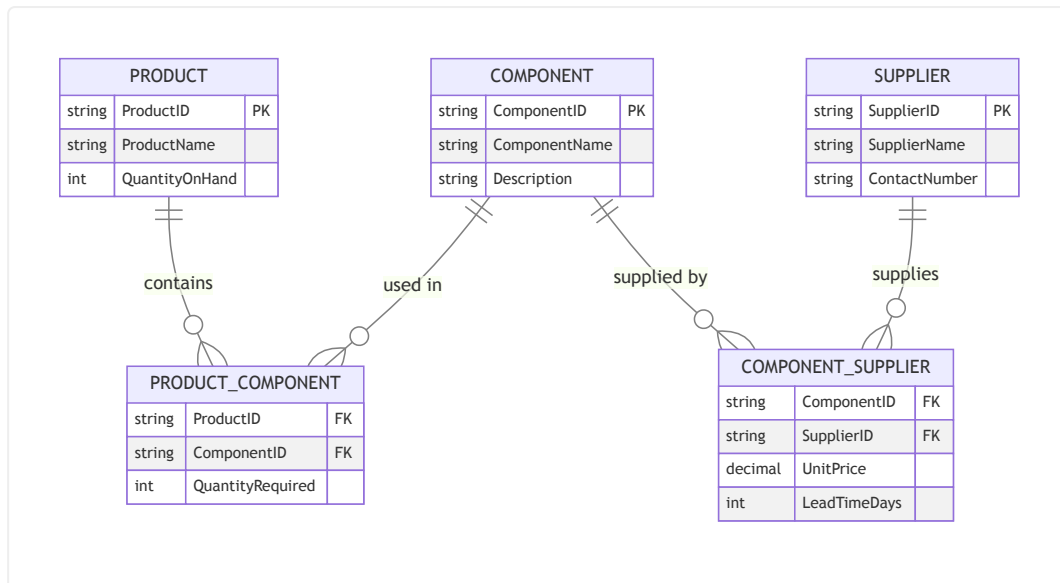
Question 6: A manufacturing company produces products.

A manufacturing company's database requires tracking products, their constituent components, and the suppliers that distribute those components.

i. ERD Cardinality and Schema Mapping

- **PRODUCT to COMPONENT (\$M:N\$):** A product is made up of many components, and a single component can be used in many products. This is resolved by the associative entity **PRODUCT_COMPONENT** (Bill of Materials), using a composite primary key (ProductID, ComponentID) and tracking the attribute QuantityRequired.
- **COMPONENT to SUPPLIER (\$M:N\$):** Each component is supplied by one or more suppliers, and each supplier distributes one or more components. This is resolved by the associative entity **COMPONENT_SUPPLIER**, using a composite primary key (ComponentID, SupplierID) and tracking UnitPrice and LeadTimeDays.

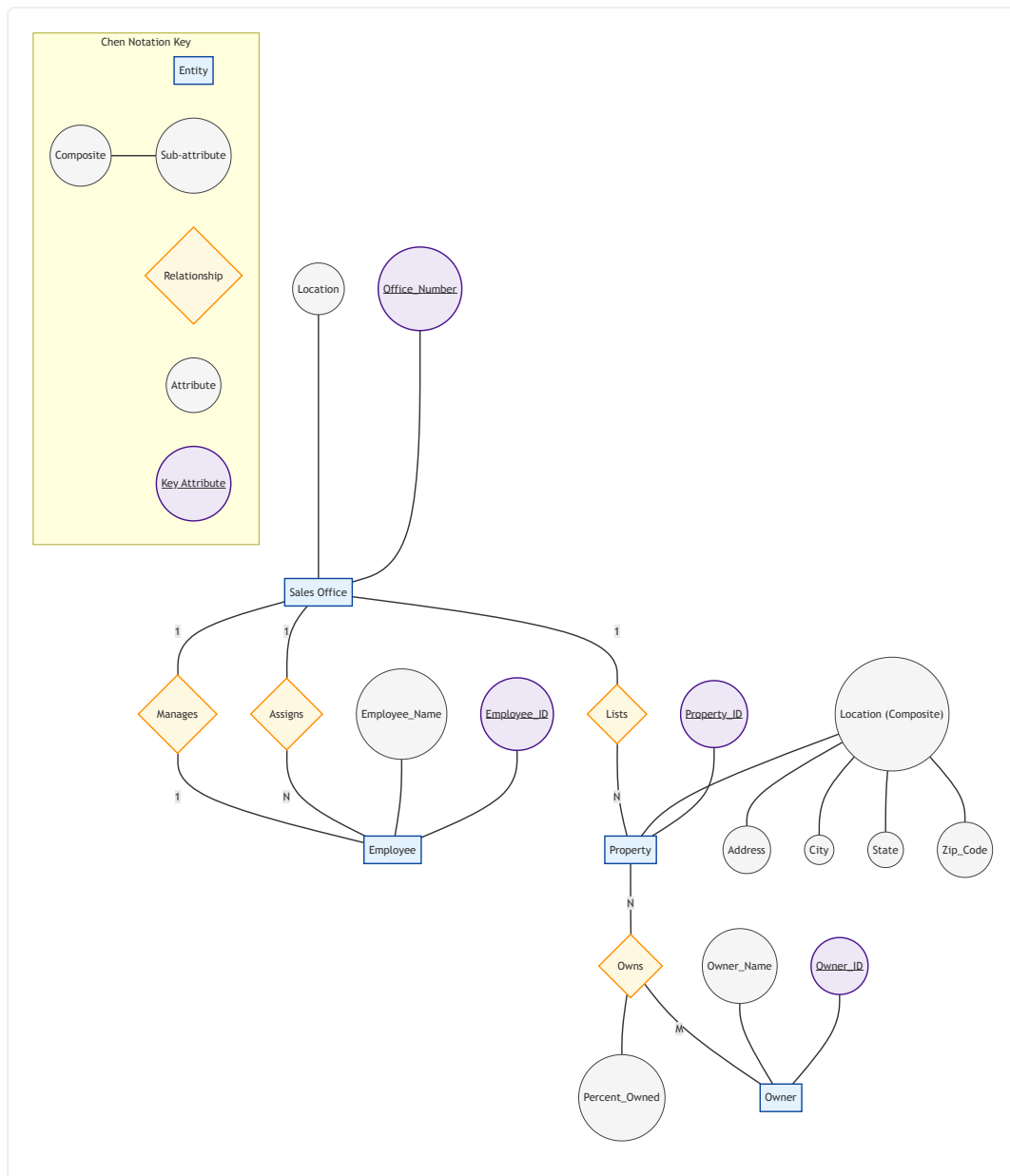
ii. Entity Names, Primary Keys, Attributes, and Relationships



Question 7: Create an ERD diagram using Chen's notation for the scenario given below.

Chen's notation uses specific shapes to represent components of a conceptual schema: rectangles for entity types, diamonds for relationship types, and ovals for attributes. Cardinalities are marked directly on the lines connecting entities to the relationship diamonds. For the real estate firm scenario, the constraints are defined as follows:

- **Sales Office & Employee:** There is a 1:N relationship (Assigns) where an office must have one or more employees, and an employee is assigned to exactly one office. There is also a 1:1 relationship (Manages) where each office must have exactly one manager, and an employee can manage at most one office (only their assigned office).
- **Sales Office & Property:** There is a 1:N relationship (Lists) where an office lists zero or many properties, and a property must be listed with exactly one office.
- **Property & Owner:** There is an M:N relationship (Owns) where a property must have one or more owners, and an owner must own one or more properties. The relationship contains the attribute Percent_Owned.
- **Property Location Composite Attribute:** The attribute Location for PROPERTY is composite, branching into sub-attributes: Address, City, State, and Zip_Code.



Notation Legend

The symbols and layout elements in the Chen's notation diagram represent specific logical constraints:

Graphical Symbol	Meaning	Example in Diagram	Description
Rectangle	Entity Type	Sales Office, Employee, Property, Owner	A primary physical or abstract object stored in the database.
Diamond	Relationship Type	Assigns, Manages, Lists, Owns	A semantic association between two participating entities.
Oval (Regular)	Simple Attribute	Location, Employee_Name, Owner_Name	A basic property or characteristic describing an entity.
Oval (Highlighted, Underlined)	Key Attribute (Primary Key)	Office_Number, Employee_ID, Property_ID, Owner_ID	A unique identifier that guarantees tuple uniqueness.
Branched Ovals	Composite Attribute	Location → Address, City, State, Zip_Code	An attribute composed of several smaller sub-attributes.
Oval on Relationship	Relationship Attribute	Percent_Owned on Owns	An attribute that belongs to the association itself, rather than either entity.
Connecting Line Labels	Cardinality Ratios	1 (One), N / M (Many)	Defines the numerical limits (1:1, 1:M, M:N) of participating instances.

References

1. Chen, P. P. S. (1976). The entity-relationship model—toward a unified view of data. *ACM Transactions on Database Systems (TODS)*, 1(1), 9–36.
<https://doi.org/10.1145/320434.320440>
2. Coronel, C., & Morris, S. (2018). *Database Systems: Design, Implementation, and Management* (13th ed.). Cengage Learning.
3. Elmasri, R., & Navathe, S. B. (2017). *Fundamentals of Database Systems* (7th ed.). Pearson.